

Networking Essential - Learning note (Day2)

Date: 18/11/2025

1. Summary of what I learned

I reviewed the fundamental concepts of computer networking and how data moves across the internet. Key points include:

- How IP addressing works(Classes of IPv4 addresses - A,B,C,D and E. And the categories - Public and Private)
- Basics of subnetting and why networks are divided into smaller segments
- Understanding layer 2 Mac addresses and layer 3 Ip addresses.
- How routers and switches operate in a network and how they aid communications.
- The difference between the OSI model(7 layers) and the TCP/IP model (4 layers)
- The Purpose of common ports and protocols (HTTP,DNS,SSH,FTP,SMTP etc.).
- How packets are encapsulated and travel from one device to another.

Networking is the backbone of cybersecurity, because every attack,threat, or incident occurs on a network.

2. Top Protocols and What They Do

Port	Protocol	Purpose
80	HTTP	Web browsing (Unencrypted)
443	HTTPS	Web browsing (encrypted)
21/22	FTP	File transfer
53	DNS	Converts domain name to ip address
25	SMTP	Sending emails
110	POP3	Retrieves email and delete the email from the server
143	IMAP	Retrieves email and keeps master copy on the server

3389	RDP	Remote desktop login
22	SSH	Remote login and command execution(Secured)
23	TELNET	Remote login and command execution(unsecured)
445	SMB	Enables files and resources sharing between nodes on a network.
123	NTP	Used to synchronize computer clock on a network

3. Quick Explanations (My Own Words)

- **How DNS Works(Short explanation)**

Domain name system (DNS) is like a dictionary that maps a domain name to an ip address. When you type a website name (e.g facebook.com), DNS converts the domain name to an IP address so your browser can reach the correct server.

- **What a packet is**

A packet is a small unit of data. Anything sent across the internet (images, videos, files, messages) are broken down into packets, transmitted, and and reassembled at the destination.

- **Switch vs Router**

A switch connects devices in a Local Area network(LAN). It forwards data frames based on layer 2 MAC addresses. It builds and maintains a MAC address table to efficiently deliver data frames to the right device. By default, a layer 2 switch creates a broadcast domain, but with the use of VLANs, it can segment a network into multiple broadcast domains.

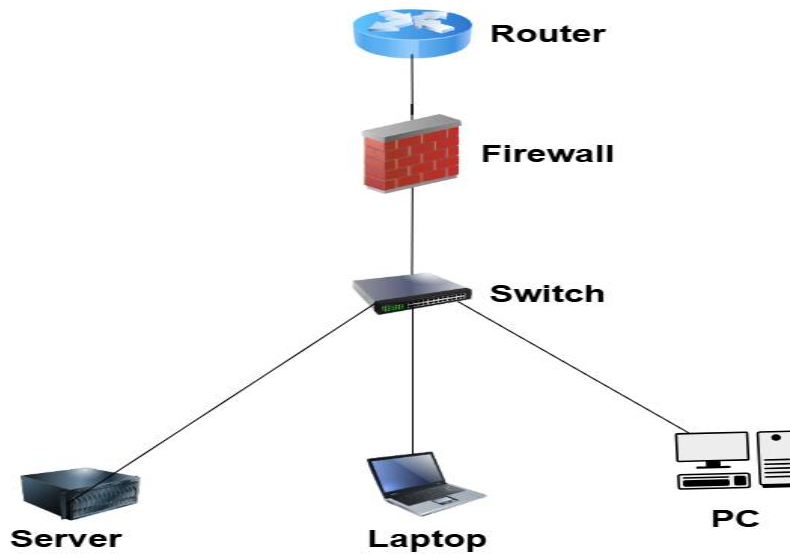
A Router connects multiple networks together. It forwards data packets based on layer 3 IP address. It determines the best path for sending data packets using ip addresses and routing tables.

4. Mini - Projects Completed Today

- **Project 1: Network Diagram Creation**

I created a simple network diagram using [Draw.io](https://draw.io) to show how basic networking devices connect inside a small network. The diagram Includes:

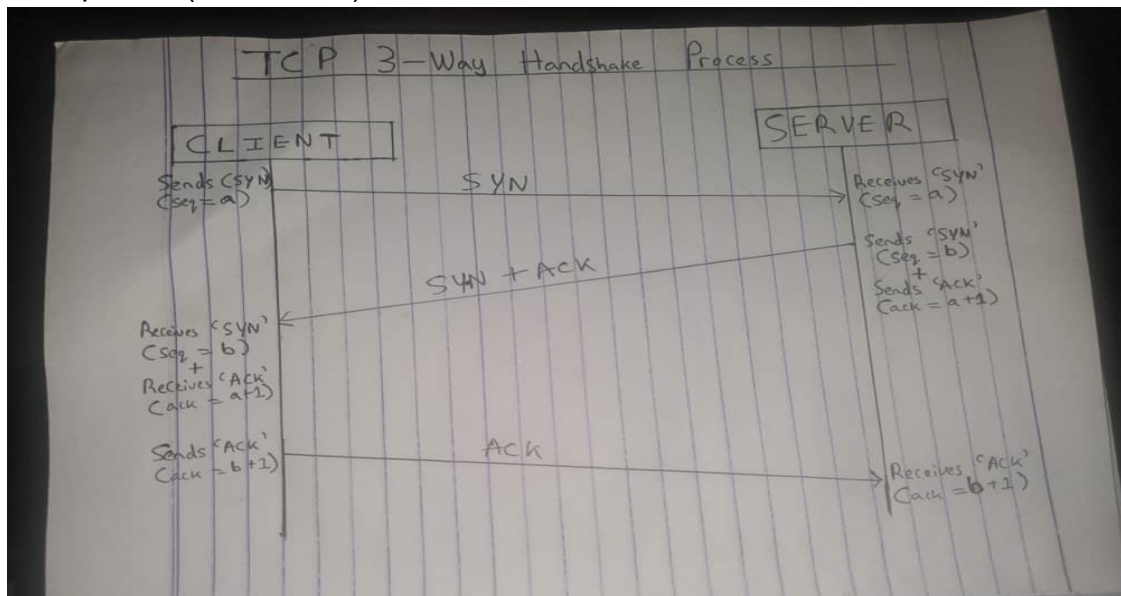
1. Users
2. Switch
3. Router
4. Firewall
5. Server



- **Project 2: TCP 3-Way Handshake Diagram**

A hand drawn representation of the TCP connection establishment process showing:

- SYN packet(Seq = a) from Client > Server
- SYN/ACK packet(Seq = b, ACK = a+1) from Server > Client
- ACK packet (ACK = b+1) from Client > Server



● Project 3: 20 Common Ports List

I documented a list of important ports used in cybersecurity investigation

Common Port	
Port 20 (TCP)	→ FTP Data (Used for actual transfer of files during FTP session)
Port 21 (TCP)	→ FTP Control (Used for control and control signal (login, logout, commands))
Port 22 (TCP)	→ SSH (Used for Secure, encrypted remote login and command execution)
Port 23 (TCP)	→ Telnet (Unencrypted remote login service)
Port 25 (TCP)	→ SMTP (Used to send email between mail servers)
Port 53 (TCP/UDP)	→ DNS (Translate domain name to IP address)
Port 67 (UDP)	→ DHCP (Automatically assign IP address and network configuration)
Port 68 (UDP)	→ DHCP (Client)
Port 80 (TCP)	→ HTTP (Used for unencrypted web browsing)
Port 110 (TCP)	→ POP3 (Used by email client to retrieve email from mail server)
Port 123 (UDP)	→ NTP (Used to synchronize computer clocks over a network)
Port 143 (TCP)	→ IMAP (Used by email client to manage and retrieve email from mail server)
Port 389 (TCP/UDP)	→ LDAP (Used for accessing and managing directory information services)
Port 139 (TCP)	→ NetBIOS (SMB over NetBIOS) (Used for file/print sharing, but relies on the older, less efficient NetBIOS layer)
Port 445 (TCP)	→ SMB (Server Message Block) (Used for providing shared access to files, printers, serial and other resources between nodes on a network)
Port 443 (TCP)	→ HTTPS (The Secure and encrypted version of HTTP, using SSL/TLS encryption)
Port 636 (TCP)	→ LDAPS
Port 514 (UDP)	→ Syslog
Port 3389 (TCP)	→ RDP
Port 993 (TCP)	→ IMAP over SSL
Port 995 (TCP)	→ POP3 over SSL
Port 161 (UDP)	→ SNMP Agent listening port
Port 162 (UDP)	→ SNMP manager listening port

5. Why Networking Matters For SOC Analysts

A SOC analyst must understand networking because attackers exploit network systems. Networking knowledge helps SOC analyst:

- Identify suspicious traffic in SIEM alerts
- Understand how malware communicates (C2 traffic)
- Investigate DDoS, Portscanning, MITM and DNS poisoning.
- Analyze Logs from firewalls, routers, VPNs, web servers, and endpoints.
- Detect anomalies like unusual ports, traffic spikes, or strange IP addresses.

Without networking fundamentals, a SOC analyst cannot interpret logs or detect threats effectively.

6. Five Questions I Still Have.

- How does a SOC analyst differentiate between normal and abnormal DNS traffic?
- Which network protocols are most commonly used in cyberattacks?
- How does VLAN segmentation help prevent lateral movement?
- What tools do SOC analysts use to analyze packet captures besides Wireshark?

- How does encrypted traffic inspection work on enterprise firewalls?

7. What I will Improve or Study Next

- Explore common attack techniques that abuse ports and protocols
- Learn deeper subnetting
- Practice analyzing more packet captures